The IEEE International Conference on
Industrial Informatics (INDIN 2008)
DCC, Daejeon, Korea July 13-16, 2008

Low Delay Live Video Streaming System for Interactive Use

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Abstract—

With the rapid growth of the Internet, users are increasingly seeking real-time multimedia application services. To meet this requirement, many streaming methods are developed. However, most of streaming systems are used for ornamental use such as video on demand(VOD). Users for such application do not require interaction, so reducing jitter is the most important issues. Therefore, users don't demand low delay systems.

On the other hand, the systems for the interactive use such as remote control, tele-surgery and so on, demand the extremely low delay and low jitter. This article propose the new live video streaming system featured by the low delay and low jitter. This paper expresses the methods of encoding and transmitting videos and describes the results of the simulations that indicate the availability of the system.

I. Introduction

These days live video streaming technology is used for video conference and video surveillance, real-time distribution, etc. and there are many ways to encode and transmit data. However, in almost all of them the image quality takes priority over the playback delay because their systems are made for ornamental use. On the other hand, the systems for interactive use, which means the operations with received feedback picture such as remote control, tele-surgery and so on, need low delay and low jitter i.e. constant delay.

The delay factors of live video streaming is as follows: *capturing delay, display delay, encoding and decoding delay, transmission delay* and *buffering delay*. *capturing delay* and *display delay* depends above all on the performance of the used camera, graphic card and display, etc. On the other hand, the improvement of the encoding and transmitting way can diminish *encoding and decoding delay, transmission delay, buffering delay*.

The problems on the road to low delay and low jitter are packet loss and delay fluctuations. These are unavoidable on the Internet. Many conventional systems use buffering in order to settle them. Buffering is exactly the useful way to avoid playback break and the image error because of packet loss and delay fluctuations. However, Buffering increases playback delay, so it should be small to realize low delay. In order to display images with the short buffering time, the encoding and transmitting way must have the tolerance strongly for packet loss and delay fluctuations.

For the above reason, we decided to develop the our own live video streaming system with low delay and low jitter. This system is for the use like remote control.

The paper structure is as follows. Next section exhibits proposed system and there are some simulations about the performance of the system in section 3. Finally, section 4 contains our conclusion and future works.

II. Related Works

Currently, the general standard for video compression is H.264/AVC[1]. H.264/AVC is the hide-compression standard. H.264 can realize twice more than MPEG-2[2], the existing standard.

The principle of the compression algorithm is basically similar to past MPEG-1, MPEG-2, H.263[3], and MPEG-4, etc. The space conversion, the inter-frame prediction, a quantization, and an entropy encoding are adopted for this compression. H.264 gets a lot of improvements to these tools and the new tools such as the arithmetic coding and filters. In addition, because several modes are properly used according to the image feature, the compression is achieved surpassing the existing standard.

However, H.264 standard consists of the extremely complex codings, so tends to give heavy load to CPU and the computation takes much time. Therefore, H.264 goes against the real-time streaming system.

Live video streaming system also becomes popular and is used for video conference and video surveillance, real-time distribution, etc. Especially, Skype[4] is the popular service of the live video system. Skype offers users the video communication with small cost. Its playback delay is small enough for usual use such as TV phone, so many people use it. However, the playback delay becomes more than 500ms according to circumstances. It is too large for the interactive use.

In order to make better live streaming system, many researches about the live video streaming are done. Most of them tackles the delay from the view point of transport protocols such as [5][6], but few article which tackles it from the view point of the coding and transmitting method. In addition, The streaming systems whose chief purpose is the low delay and jitter are quite few.

III. Proposed System

The fundamental design principle to avoid low delay is that one side called *server* sends a frame data soon after the camera captures it, and another side called *client* displays it soon after receiving it. If we can implement this in all frames at the same interval, low delay and low jitter system is realized. However, in the real there are some problems as follows.

Encoding complex and bandwidth constraint

When we send data in restricted bandwidth as soon as possible, it is an efficient way to encode the data and bring their size down. However, encoding and decoding time are also the delay factor in live video streaming system. The relationship between the compression ratio and the encoding time is usually trade-off,

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i.e. if we compress data with high ratio in order to shorten the transmission delay, the encoding delay may increase. We need to select the algorithm minimizing the following formula.

$$min\ (EncodingTime + \frac{DataSize}{Bandwidth})$$

Packet loss over the Internet

Packet loss is generated all over networks. Live video streaming system generally resends or abandons the frame when its packet is lost. However, proposed system should not resend the lost data. The transmission delay is large in delay factors, so re-sending it makes the total delay much large. In addition, the resent data might be too late when they are received in tight stale demand. Therefore, it is necessary for our system not to resend the lost frame but minimize the influence of packet loss. When seen from this point of view, the compression coding methods like MPEG and H.264/AVC are unpreferable for this system, because they compress data efficiently by using difference information between frames, i.e., one frame error has an influence on other frames referring to it.

Delay fluctuations due to the network condition

The transmission bandwidth is not always constant. All data may not be received by the scheduled time because of drastic bandwidth narrowing. Then, the system should not wait for the time receiving all frames data because the playback delay becomes longer. On the other hand, we should avoid skipping the display of the error frame because it degrades the operational performance of interactive use. Therefore, the desirable way to keep the delay and the jitter show is to partly update display by using only received data though all the data which belongs to a frame were not received by its scheduled time.

We proposed the live video streaming system based on the above considerations.

A. System Overview

Proposed system divides a frame into some blocks, and compresses each block using JPEG. In addition, the system calculates the difference between each block and the previous frame block as the amount of change. This system can diminish the amount of transmitting data because it does not send the block if the change of it is small(the amount of change is less than the threshold).

The speed of this encoding method is almost as fast as that of Motion JPEG, and its transmission is faster than that because the amount of transmitting data is smaller. In addition, the influence of packet loss can keep small in this because each encoded block is independent to other blocks and can be decoded from itself. One packet loss has a influence only on a block, and re-sending is negligible.

The frame division has one more advantage that has tolerance for the delay fluctuations. All the blocks can be independently decoded, so this system can partly update its display image by using only received blocks although all blocks of the frame were not received by the scheduled time because of the delay fluctuations. Therefore, this live streaming system can keep the delay constant(low jitter).

B. System Detail

Compression Coding of each divided block

Each block is encoded as standard JPEG code, i.e. the system executes discrete cosine transform(DCT) and scalar quantization, run length coding, Huffman coding, etc.

Therefore, the sum of the blocks data is almost the same amount as the frame data compressed by standard JPEG.

However, there are some differences between JPEG and the proposed system. The standard JPEG add the tables of Huffman coding, etc. as the header to the image data. On the other hand, this system sends these tables in advance because these data are common for all blocks, so it is not necessary to add them to the individual image blocks.

Data Transmission

This system uses both TCP and UDP for data transmission, and choose them for any purpose. Before frame data transmission, the system needs to send the necessary information for video streaming. The system sends the information about the resolution and refresh rate of the camera, Huffman coding table, etc. by using TCP. For these data, it is necessary to use the reliable connection because of the data importance and not necessary to transmit fast because of the transmission before the start of video streaming. On the other hand, this system uses UDP to send the frame data. This system places a lot of emphasis on low delay, so the protocol with higher transfer rate was chosen.

For these reason, the connection between the server and the client is established as Fig.1.

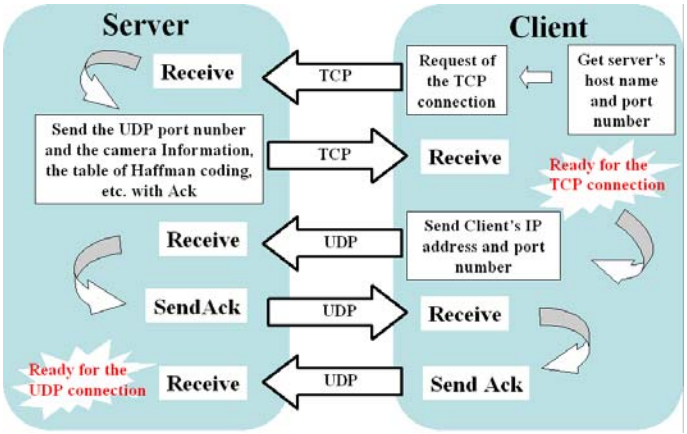

Fig. 1. The establishment of the connection

Transmission priority

This system calculates the transmission priority for each block, and sends the blocks in descending order of the priority. The arbitrary value is given as the threshold for the priority, the system does not send the blocks whose priority is lower than the threshold.

The the transmission priority is given by two elements: the difference of the image between frames and the age of blocks. This system was designed based on the idea that the blocks whose visual change is larger are more important and should be

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updated. In contrast, the blocks with small change should not be updated because it is negligible enough. So, this system adopted the difference between the previous frame and the current frame as the element of the transmission priority.

However, to decide the priority only by the deference is not good choice, because it has the possibilities that some unrenewed blocks at the client. If the change of some blocks are constantly small, they are always low priority and can not be updated forever. It is undesirable. Therefore, in order to avoid this problem , the system adds the age (the information how old each block is) to the evaluation for the transmission order.

In this system, all blocks have their own age. Whenever the frame is updated, only unrenewed blocks of the frame count their own age up. Their age returns to zero when the blocks are updated, and they begin to count their age up again. Therefore, the older the block images get, the higher the value of transmission priority become, and the problem of the unrenewed blocks is settled.

The comparison function of the proposed system is expressed as follows.

$$Priority = Difference + W_{age} \times Age$$

Priority is the transmission priority, $Difference$ is the difference between frames, *Age* means how old the block is, and W_{age} is the weighting function.

The weighting function for aging is decided by user's demand. For example, when half of all the blocks in a frame are always updated, the rest half of the frame are not updated at first. However, as the age of the rest blocks becomes big, the factor of the age becomes large and they get updated little by little. Eventually the updated blocks because of both the difference and the age are generated.

The ratio between two types of update is approximately represented as a follow.

$$Update_{diff} : Update_{age} = Difference_{MAX} : W_{age} \times Age_{MAX}$$

$Update_{diff}$ means the number of update blocks by their differences, and $Update_{age}$ means that by their ages. $Difference_{MAX}$ and Age_{MAX} are each the maximum value of their differences and ages. Users can decide W_{age} according to how many blocks they want to update due to their age.

Threshold of transmission priority

This system have the threshold of transmission priority in order to save the unnecessary transmission. The transmission priority of the blocks is compared to the threshold and the system do not send the blocks whose priority is less than the threshold. The comparison to the threshold is done before encoding of the blocks, so the system can save unnecessary encoding time. This parameter is adaptable, so the user can change it according to the requirement.

Stable number of update blocks

Though the proposed system has tolerance for packet loss due to the blocking, it does not have provide the guarantee for stable update of the frame. If the stable update is not assured, the image quality at the client becomes low as packet loss increases. In order to avoid this problem, the system assures the stable update by the number of received blocks at the client. This system feeds the information at the client back to the server in order to control the number of sending blocks.

When the method to control the number of sending blocks is simplified, the formula is as a follow.

$$Sending(t+1)$$

$$= Required(t+1) + W_{stable} \times (Required(t) - Received(t))$$

$Sending$ means the number of sending blocks and *Required* is the required number of update blocks, *Received* is the number of received blocks at the client, and W_{stable} is the weighting function. Because of this feedback, the number of update blocks is assured.

Packetization

In this system, a packet conveys the data of some blocks to the client. This system packs the encoded blocks in a packet according to the transmission priority. The size of all the packets to contain blocks data is same. When the packet is filled with the data of blocks, the filled packet is sent to the client. After that, the system begins next packetization.

Time chart of proposed system

In this system, users can decide the delay requirement and the limit of playback delay changes according to it. In order to make the jitter of the frame update low, it is necessary for the processing phase at the server to equal that at the client. In addition, One more important thing to realize the system is to feed the information of the client back before the next decision of the priority. However, though the data are received in time, the server can decide the transmission priority according to the previously received information. The time chart of proposed system is shown in Fig.2.

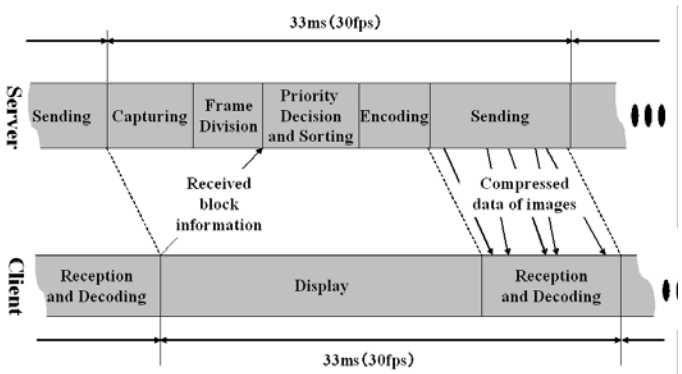

Fig. 2. The time chart of proposed system

IV. Experiments and Discussions

In this section, we evaluate the performance of the proposed live streaming system.

The PC used in the experiment has Xeon 2.8GHz, 1GB memory. The operating system is Linux 2.6.13-15-smp and we use C++ as the programming language. The camera has 640 × 448 pixels and 30fps.

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A. Influence of Packet Loss

This system can update the frame by using the data received by the update time though some blocks are lost because of packet loss.

In order to confirm that, we intentionally dropped 25% of the data at random. The image received at the client is 3.

Fig. 3. The image at random packet loss

This image was captured when the hand suddenly moved into the frame. We can found the lost blocks because of packet loss, but the system could almost keep up with the update of the frames.

B. Measurement of Delay

Second, we measured the delay related to the system performance.

We expressed that delay consists of some elements. The delay directly related to the usability of the system of them is the time lag between the real world at the server and the displayed image at the client.

However, of all the delay elements, both the capturing delay and the display delay can not be shortened by the software technology, but by the hardware performances such as the camera and the graphic board, the monitor and so on. In addition, the influence on the total delay is large. Therefore, when we evaluate the playback delay of the system, we should evaluate it without the capturing and display delay. In order to achieve it, the measurement was done only in the PC.

Measurement of playback delay

We measured the the playback delay of this system. The delay was measured by the elapsed time from the capturing to the execution of displaying. The result is shown in TABLE I.

TABLE I
THE PLAYBACK DELAY OF PROPOSED SYSTEM

Number of sample	Average time[ms]	Max. value[ms]	Min. value[ms]	Standard deviation [ms]
1000	41.36	71.83	24.18	8.86

The result indicates that the playback delay is small. In addition, it is clear that the maximum value of the delay is a little large but the standard deviation is small, so the delay is constant. This is the measurement in the local area network, so the influence of the network delay is small, but it is favorable enough that this system realizes low delay and low jitter in the global network.

Playback delay and received blocks

We examined whether the program behaved as intended by measuring the number of received blocks per cycle at several required delay.

We measured the number of received blocks at the client and the received blocks were divided into three groups: the current blocks which should be displayed in current frame, the past blocks which has been already old and not been displayed any more, and the future blocks which are necessary in the next frame or later.

In this experiment, the server always sent all the blocks in a frame. A frame divided into 300 blocks. In addition, received blocks goes to the buffer which can accommodate 300 blocks untill the next processing. The old packets are discarded if over acceptable amount according to the feature of ring buffers. Fig.4 shows the results of this experiment.

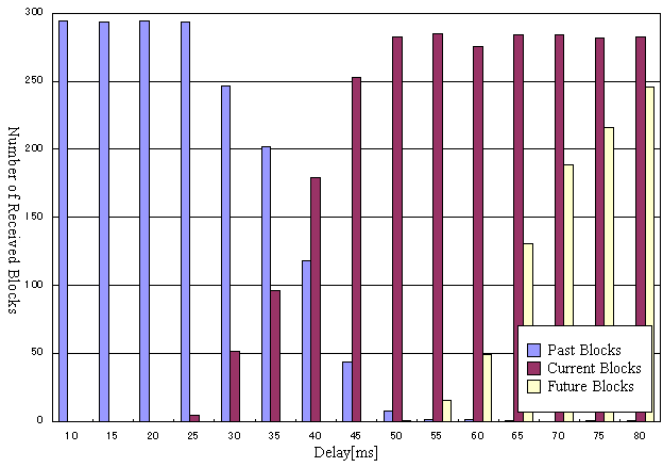

Fig. 4. The relation between the playback delay and the number of received blocks

The graph indicates that the number of current blocks becomes large with the moderation of the playback delay demand. In contrast, the number of past blocks decreases with the moderation.

This is because past blocks are the current blocks of the previous frame which could not be received in time. It is natural that the number of past blocks decease as the number of current blocks increase.

However, it is clear that setting the limit of the delay high has bad influence on the delay performance. In this case, it is optimal to set the required delay 50ms.

This experiment declared the user of this system can set the optimal delay offset according in this way.

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C. Availability of Feedback

The availability of the feedback functions were estimated here.

Comparison about image quality

As previously described, The purpose of the update stabilization is to enhance the image quality with the feedback. we evaluated whether it averts the degradation of the image quality by using SSIM(Structural Similarity)[7]. One of the most popular evaluation method for image quality is PSNR(Peak Signal-to-Noise Ratio), but it is not enough to evaluate the quality because its evaluation always does not synchronize the subjective estimation. On the other hand, SSIM evaluates directly the similarity of the images structure by using the brightness and the contrast, etc. Therefore, it needs the original image without the degradation. Compared to PSNR, SSIM estimation is said to be closer to the subjective estimation, and some H.264 encoders adopt this method.

We compared SSIM of the images between with the feedback and without the feedback. We made the required number of received blocks changed from 10 to 70 at 10 intervals and estimated the degradation of the displayed image at the server by SSIM. The weighting function for aging is 30 and that for the update stabilization is 0.5. In addition, the capturing speed got late to 3fps because the computation time of the image saving for SSIM is very large.

The simulation result is shown in Fig.5.

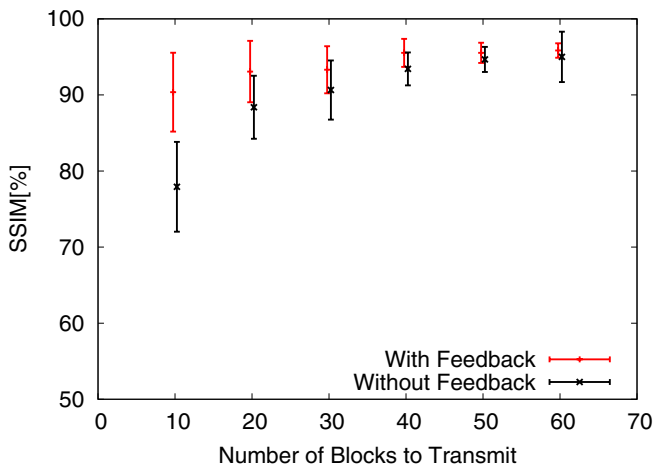

Fig. 5. SSIM with and without the feedback

In the graph, SSIM value with the feedback keep high compared to without the feedback at the small required number. This indicates that the update stabilization keeps the image quality high.

In contrast, the SSIM difference between them is small at the large required number. The reason is updating the blocks with small change increases at the large number. The block with small change has small influence on the image structure, so has small effect on SSIM. In addition, one more reason is why the feedback does not work efficiently in the vicinity of 70 blocks because a frame is made up of 70 blocks.

Aging

This system has the aging function for the transmission priority. we confirmed if the aging behaved. We evaluated the age when the waiting factor(W) changed. The evaluated value is the age of the most unrenewed block and the number of the received blocks to the client is 40.

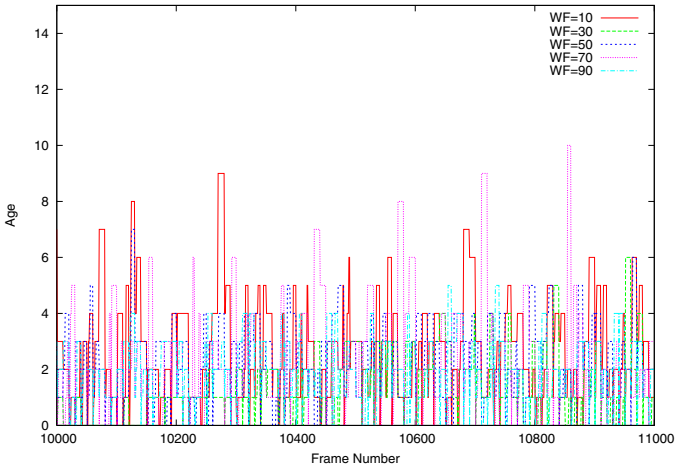

Fig. 6. The age of the most unrenewed block

Fig.6 represents the result. The transverse axis expresses a thousand processing time from 10000 to 11000. the VAR-tical axis expresses the age. We find that all the blocks kept younger than 10. This indicates that the Aging function worked efficiently.

V. Conclusion and Future Works

In this paper, we proposed the new live video streaming system with low delay and low jitter for the interactive use such as remote control, and evaluated the system performance.

The experiment showed that the proposed system had the almost equal ability to Motion-JPEG in the compression rate. In addition, all the delay due to the proposed method is comparatively low. the result showed that this live streaming system is low delay and low jitter. In addition, we could actually confirm the low delay and the low jitter and the tolerability to the delay fluctuations. In addition, the system made the image quality higher with the update stabilization and the aging function.

The future research is required to make the method of determination for some parameter and actually use the system over a lossy network.

VI. Acknowledgment

This work was supported by Grant-in-Aid for Scientific Research(C)(18500138) and (B)(20300079): matching fund subsidy from JSPS(Japan Society for the Promotion of Science).

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